Harnessmith

COMPLETE APP WALKTHROUGH

Harnessmith

Give an agent a goal. Watch it improve its own harness.

Feature Guide & Screen-by-Screen Reference · November 4, 2026

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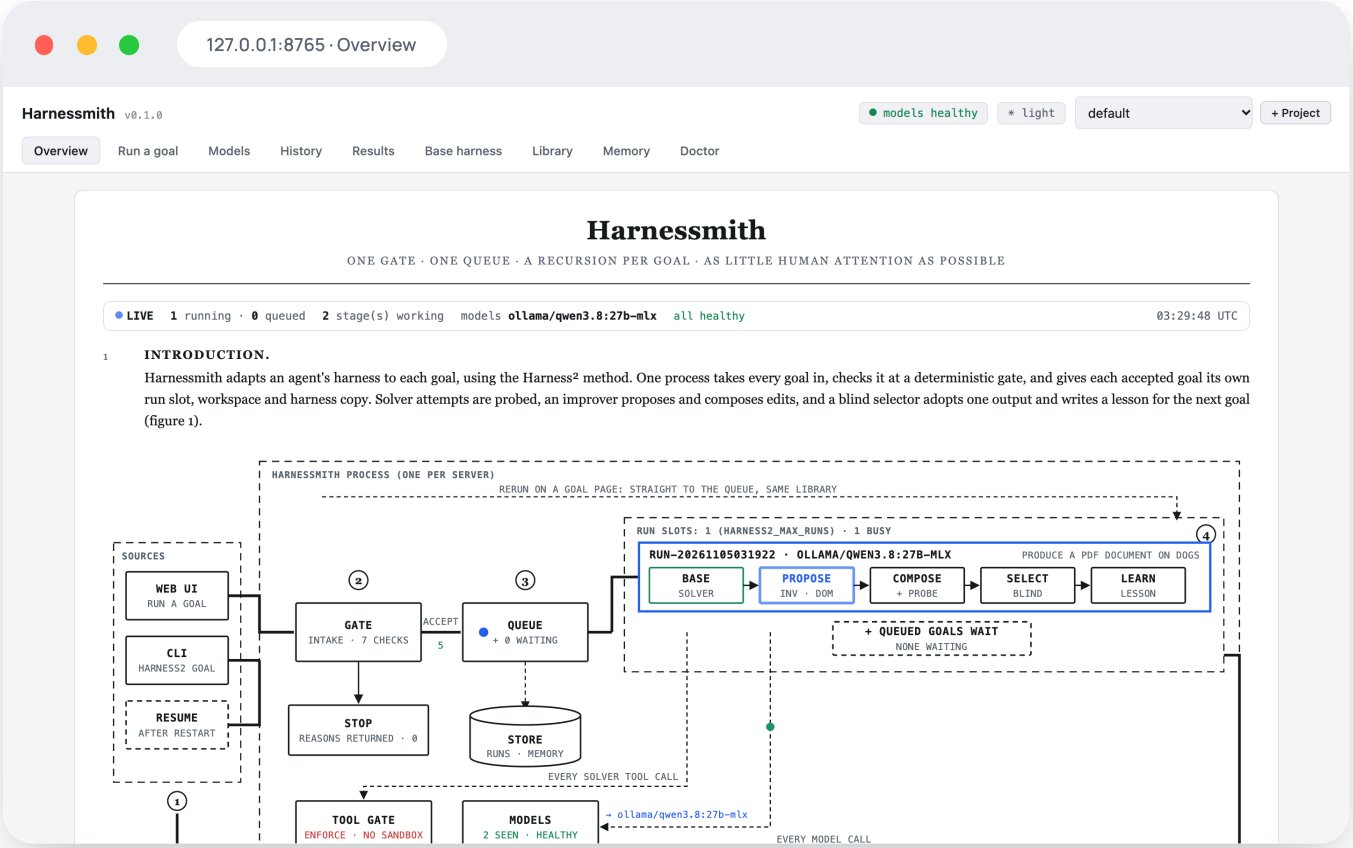
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HARNESSMITH

Agents that improve their own harness

This is Harnessmith. You give it a goal, it runs an agent on it, and then it improves the agent's harness for that goal: the instructions, rules and tools around the model. Here is a quick tour.



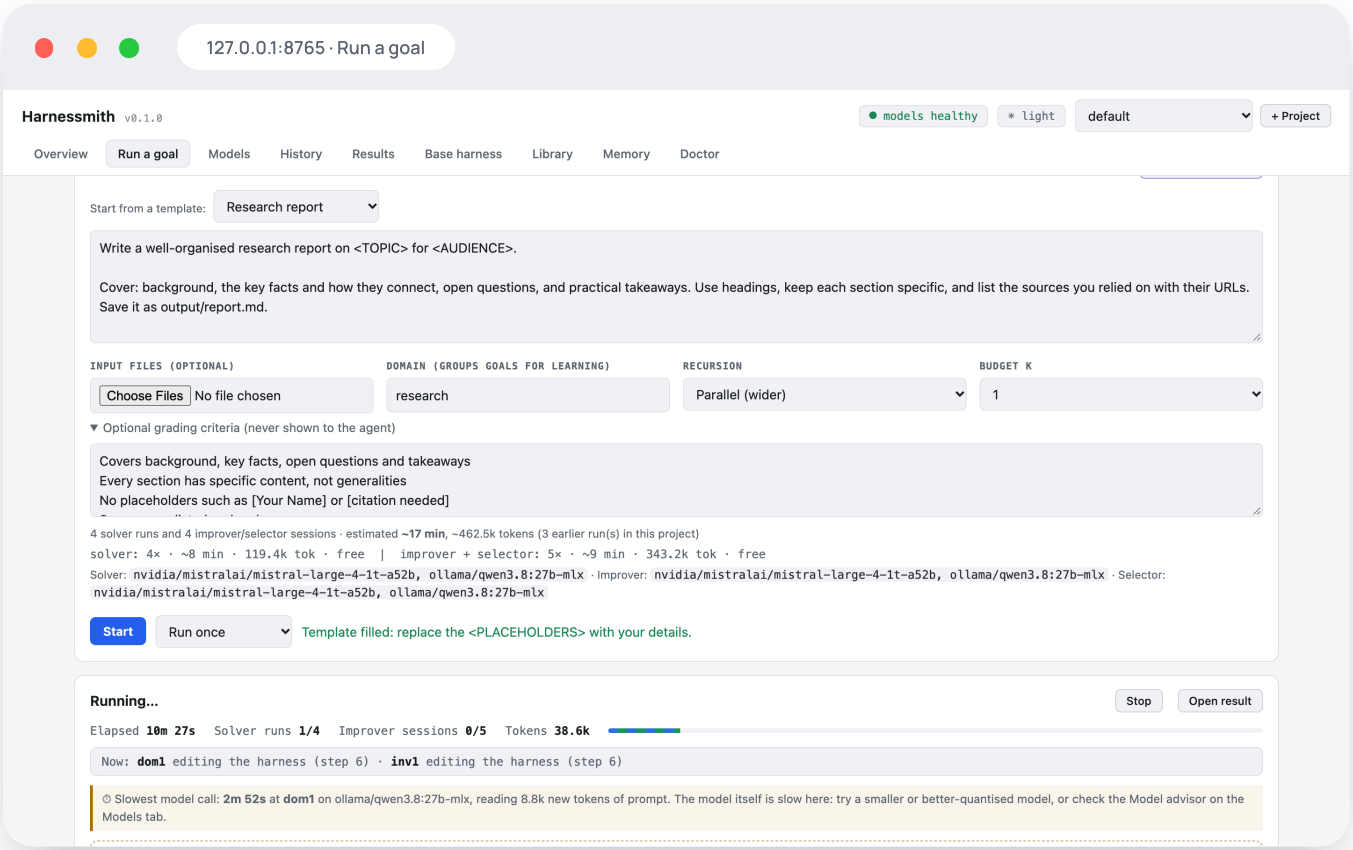
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ONE GATE, ONE QUEUE, A RECURSION PER GOAL

The whole system on one page

The Overview draws the whole system and updates live while goals run. Each goal is checked at a deterministic gate, waits in a queue, and gets its own run slot with its own workspace and copy of the harness.

- Solver attempts are probed; an improver proposes edits on two dimensions (how the agent works, what a practitioner knows) and composes them
- A blind selector picks the best output without knowing which harness made it
- A lesson is written to the library for later goals in the same domain
- The live bar shows what is running, on which model, and any model that is down



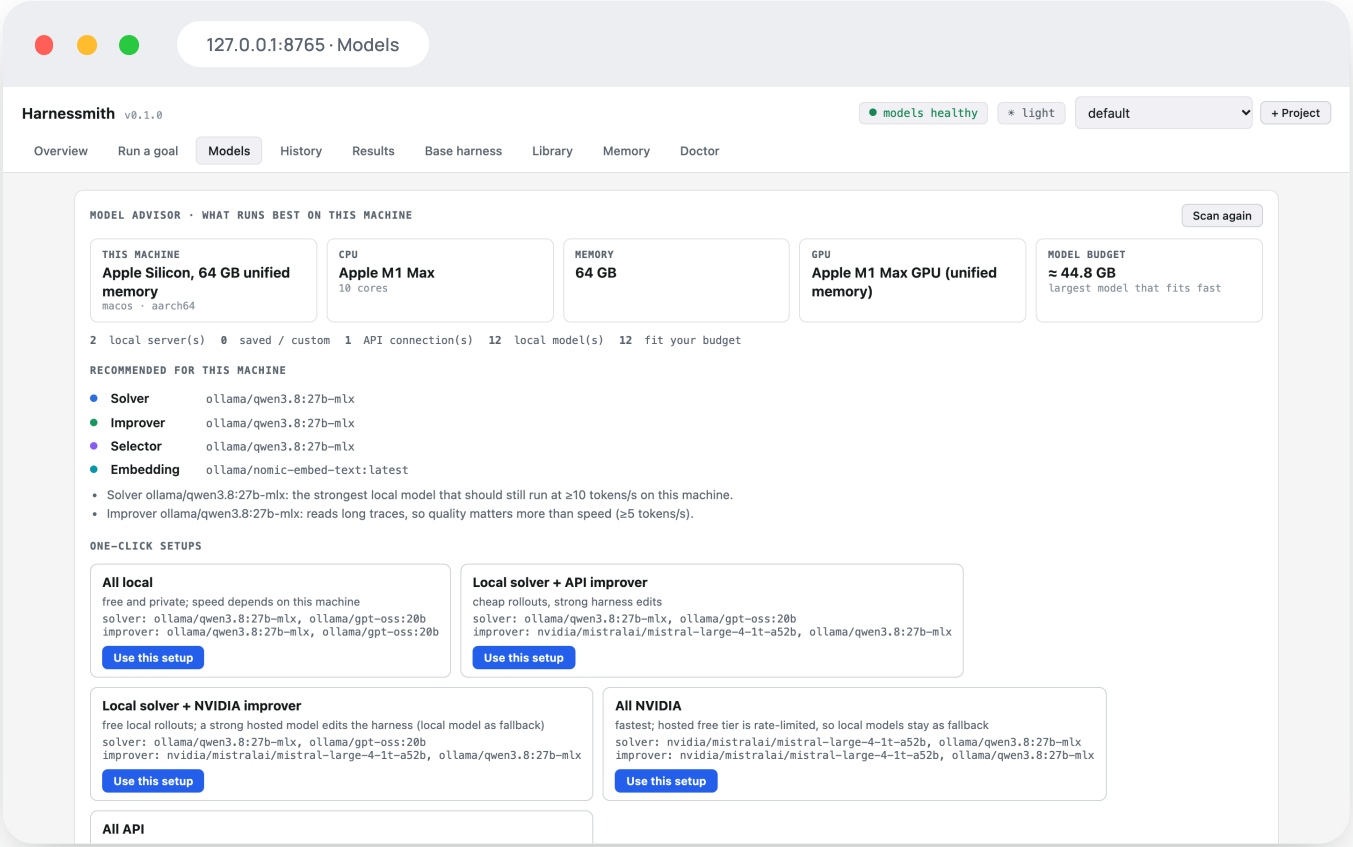
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FROM A SENTENCE TO A QUEUED RUN

Describe what you want

Describe the outcome you want, or start from a template. Grading criteria are optional but unlock scoring, A/B tests and the Results tab.

- Templates: research report, landing page, data brief, PDF document, small script
- The estimate splits runs, time, tokens and cost by role before you start
- A warning appears when the improver is a small local model, the usual weak spot
- Cmd/Ctrl+Enter starts the run; a browser notification says when it finishes



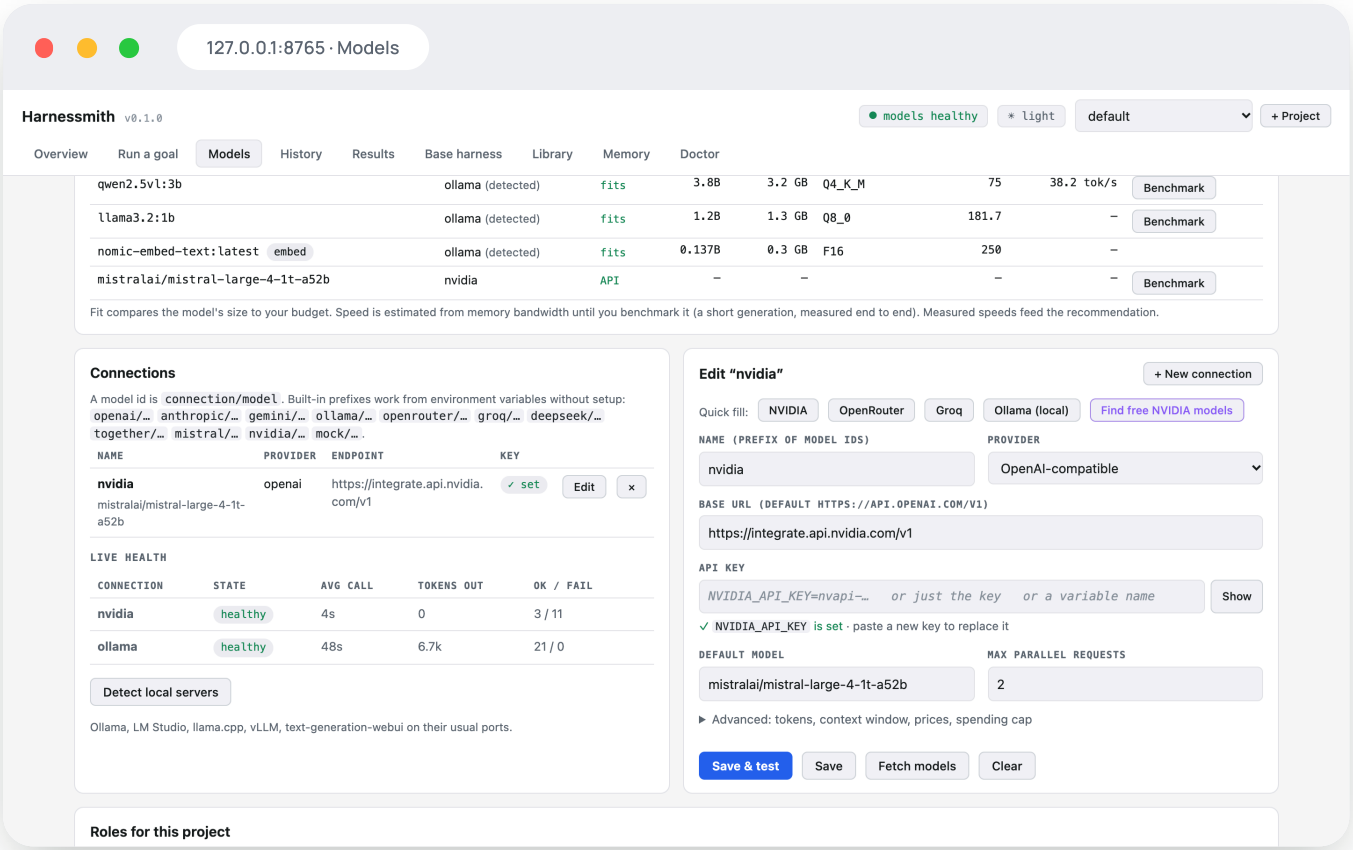
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PICK MODELS THAT FIT YOUR MACHINE

What runs best on this machine

The advisor reads your CPU, memory and GPU, lists every local model that fits, and measures real speed with Benchmark.

- Recommended solver, improver, selector and embedding model
- One-click setups: All local, Local solver + API improver, Local solver + NVIDIA improver, All NVIDIA
- Each role can list fallbacks after a comma, so a failing model hands over automatically



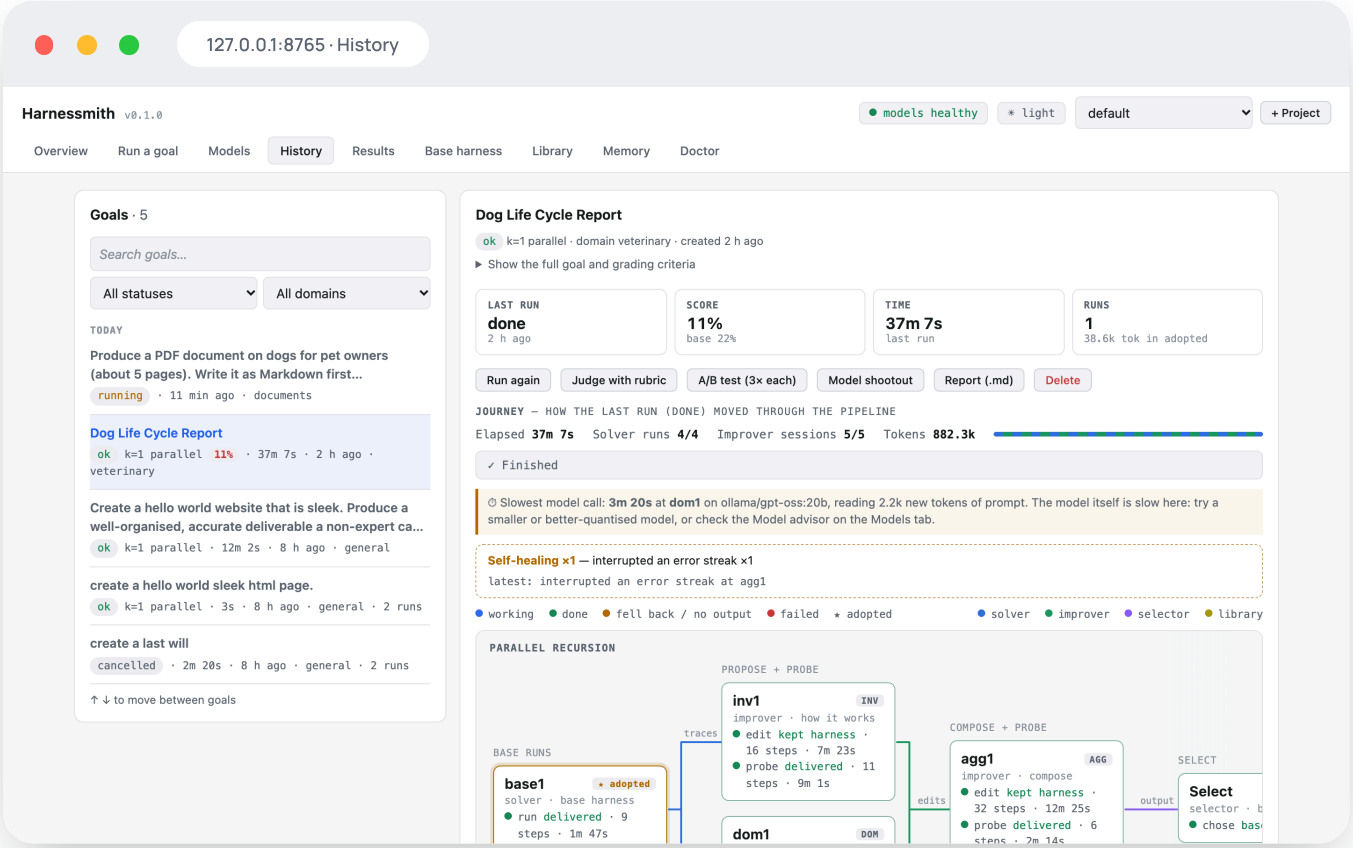
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KEYS STAY IN YOUR .ENV FILE

Connections and keys

A connection is a named endpoint such as nvidia, ollama or openrouter. Model ids are written connection/model.

- Type NAME=key (or just the key) in API key: it is written to the private, git-ignored .env file, never to connections.json
- After saving, the form only says whether the key is set; no part of the key is shown
- Save & test checks that the model really answers; Find free NVIDIA models shows which hosted models your account can use
- A model your account cannot use is skipped for the next one in the chain without blocking the connection



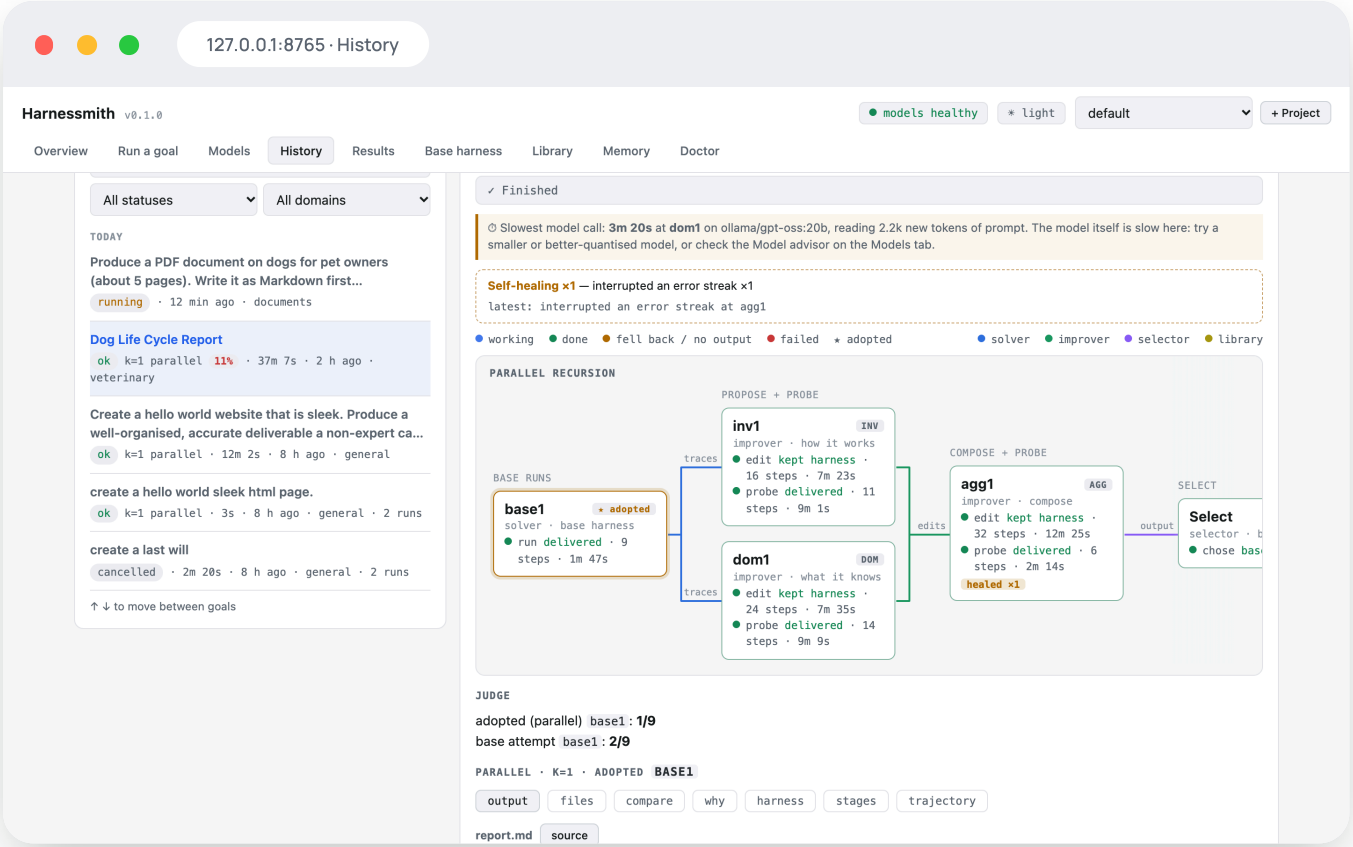
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EVERY GOAL, SEARCHABLE

Every goal, searchable

History groups goals by day. Each entry shows status, rubric score, run time and age; arrow keys move between goals and each goal has its own link.

- Search, status filter (running, needs review, done, failed, cancelled) and domain filter
- A summary card: last run, score against the base attempt, time, number of runs
- The full goal and its criteria fold away; every run of the goal is listed



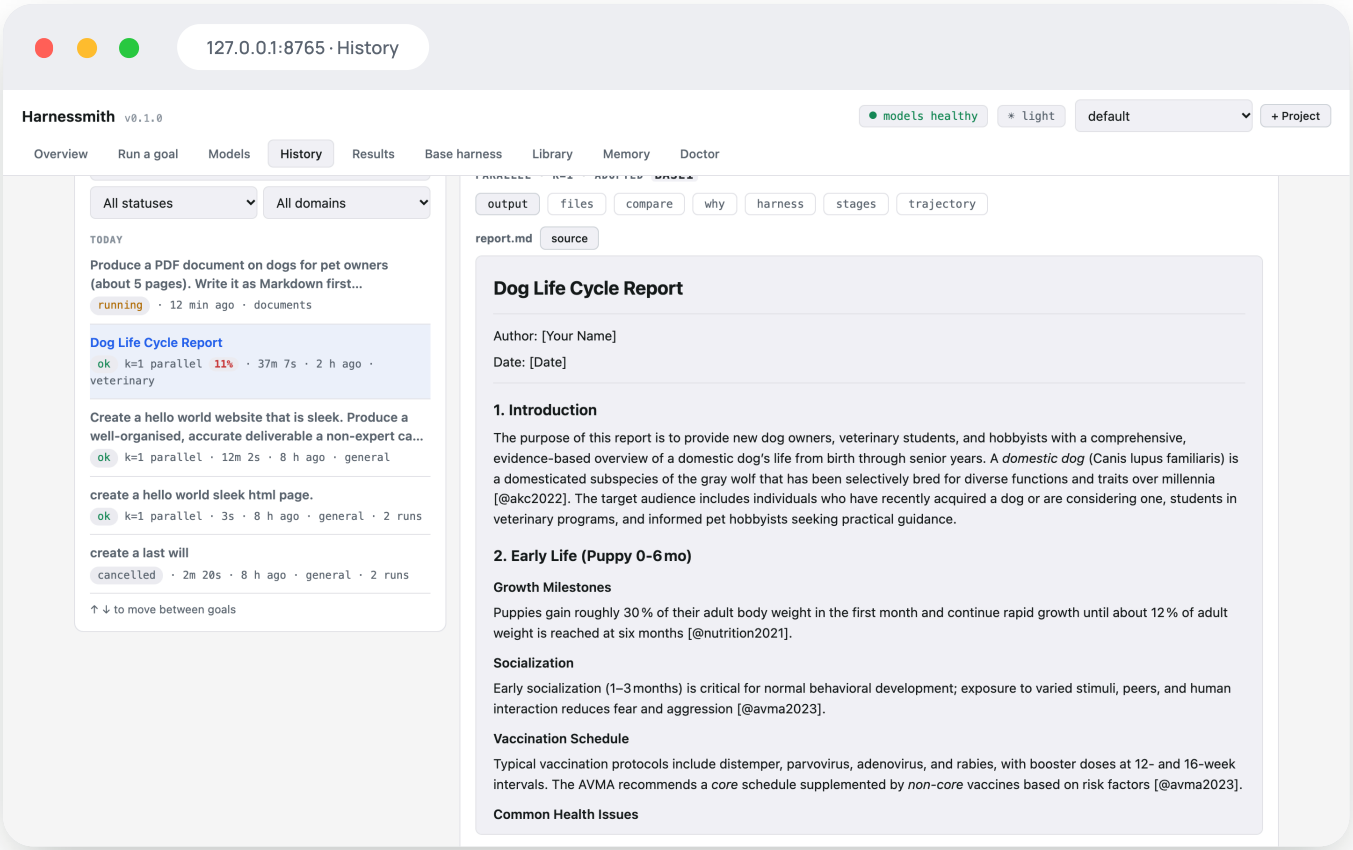
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HOW A RUN MOVED THROUGH THE PIPELINE

How the run moved

The journey map redraws the run: base runs, the INV and DOM proposals with their probes, the composed harness, and the blind selection.

- Each box shows its role, what it did, steps and time
- The adopted path is drawn as a striped trunk
- Self-heals (failovers, loop stops, rescues) are badged on the box where they happened
- Click a box for its timeline, filtered to errors and self-heals if you like



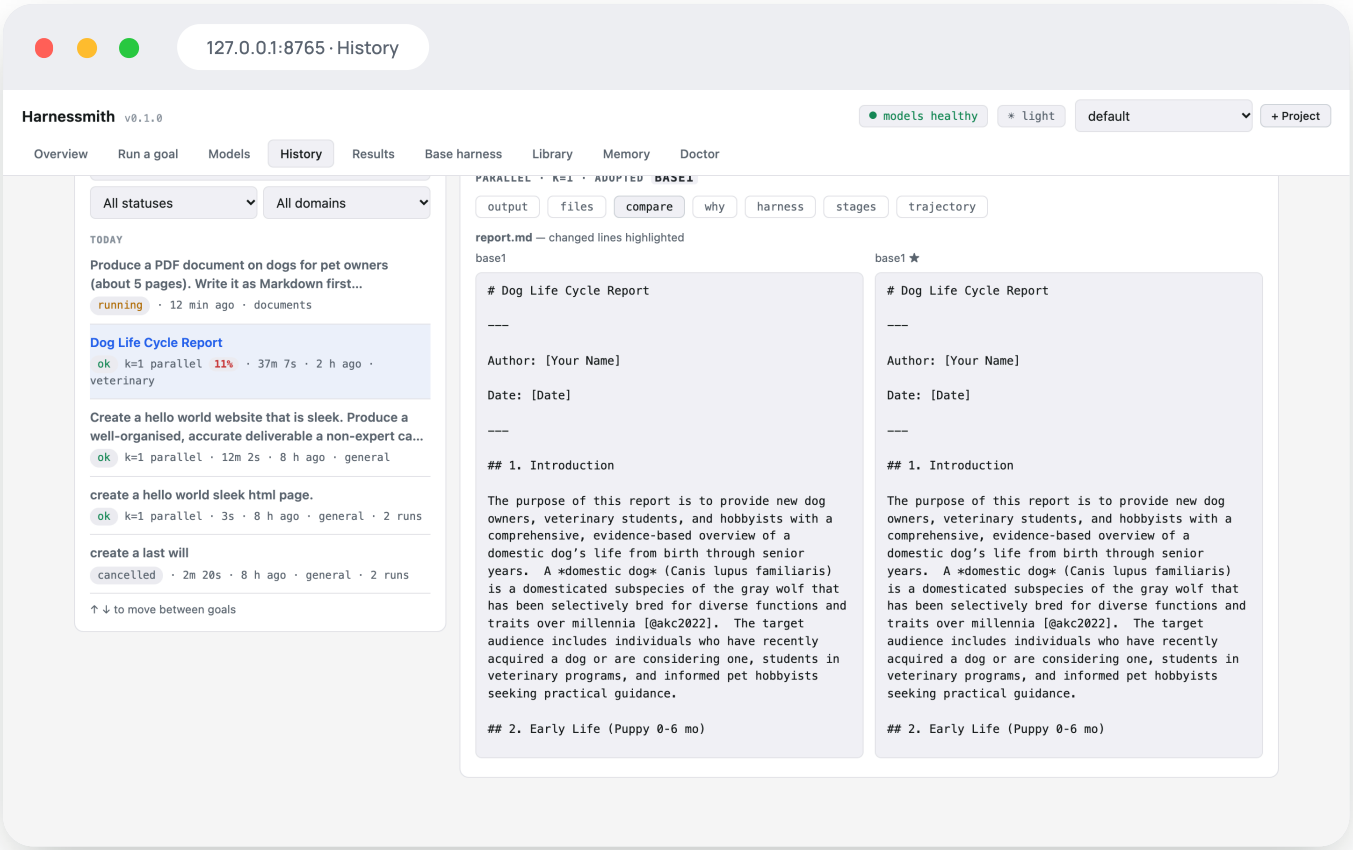
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HONEST RESULTS

The deliverable, rendered

Markdown deliverables are rendered, with a source view. The deliverable review runs on every output.

- Missing file types, fake PDFs, placeholders such as [Your Name] and citations without references are flagged
- A run whose output fails review, or scores under 50% against the rubric, ends as needs review instead of done



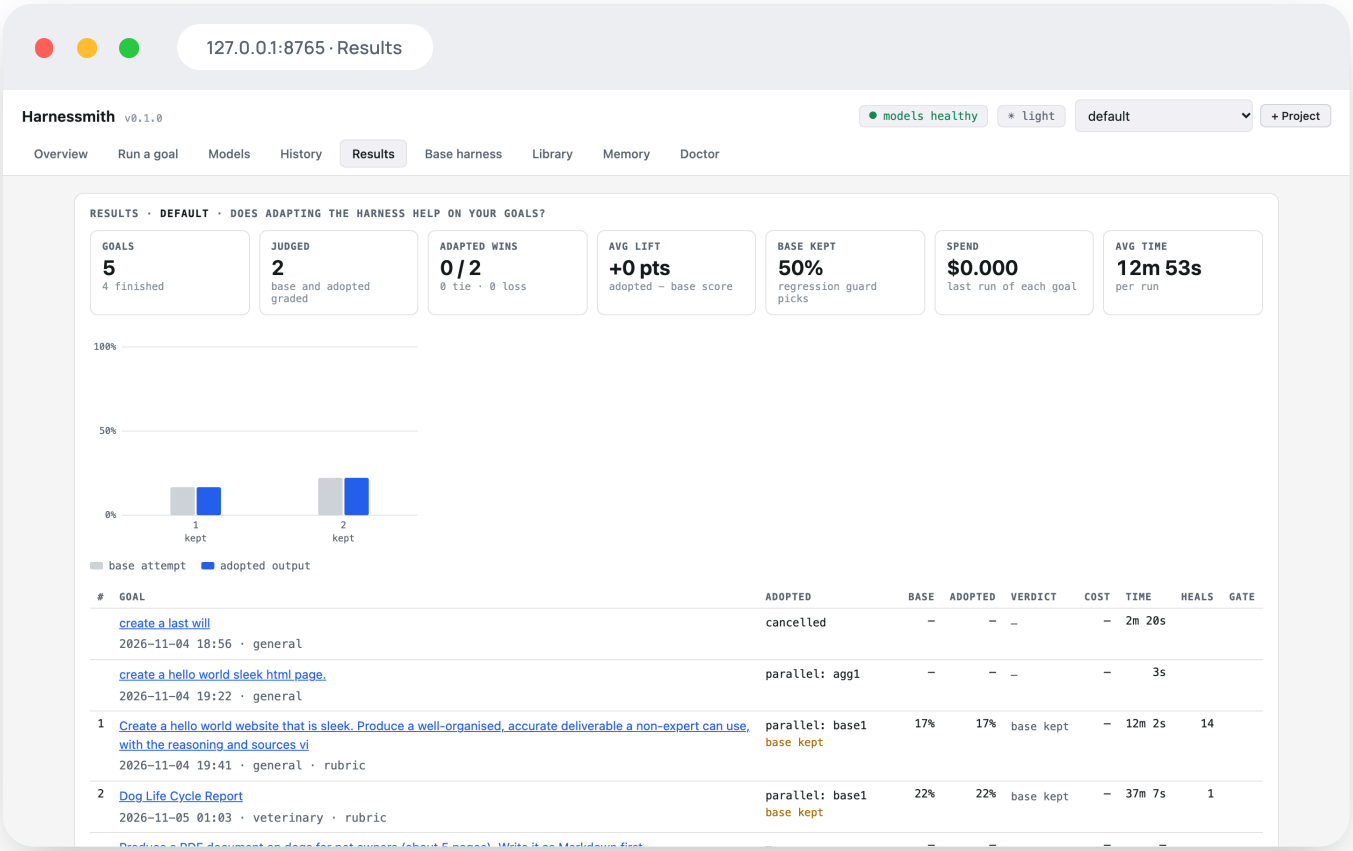
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WHAT ADAPTATION CHANGED

Base beside adopted

Compare shows the first base attempt beside the adopted output with changed lines highlighted.

- A/B test: base harness vs adopted harness, three times each, graded, with mean, spread and a verdict
- Model shootout: the same goal on several solver models, side by side
- Report (.md) downloads a shareable summary; Delete moves a goal to the project's trash

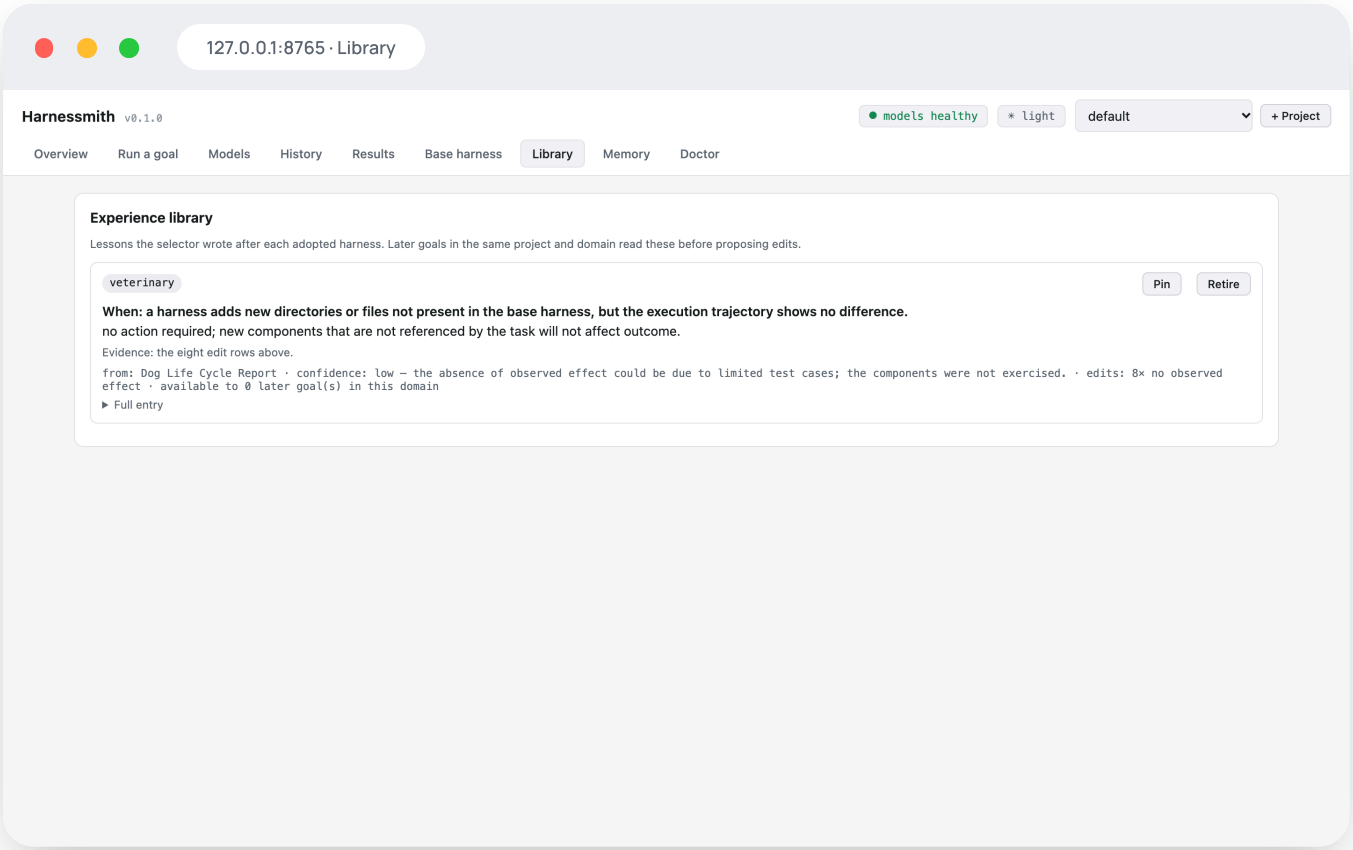


IS ADAPTING THE HARNESS HELPING?

Is adaptation helping?

For goals with criteria and a judge model, the adopted output and the unmodified base attempt are both graded after each run, three gradings each with a majority vote per criterion.

- Adapted wins, ties and losses, and the average lift in points
- How often the regression guard kept the base attempt
- Spend and time per goal

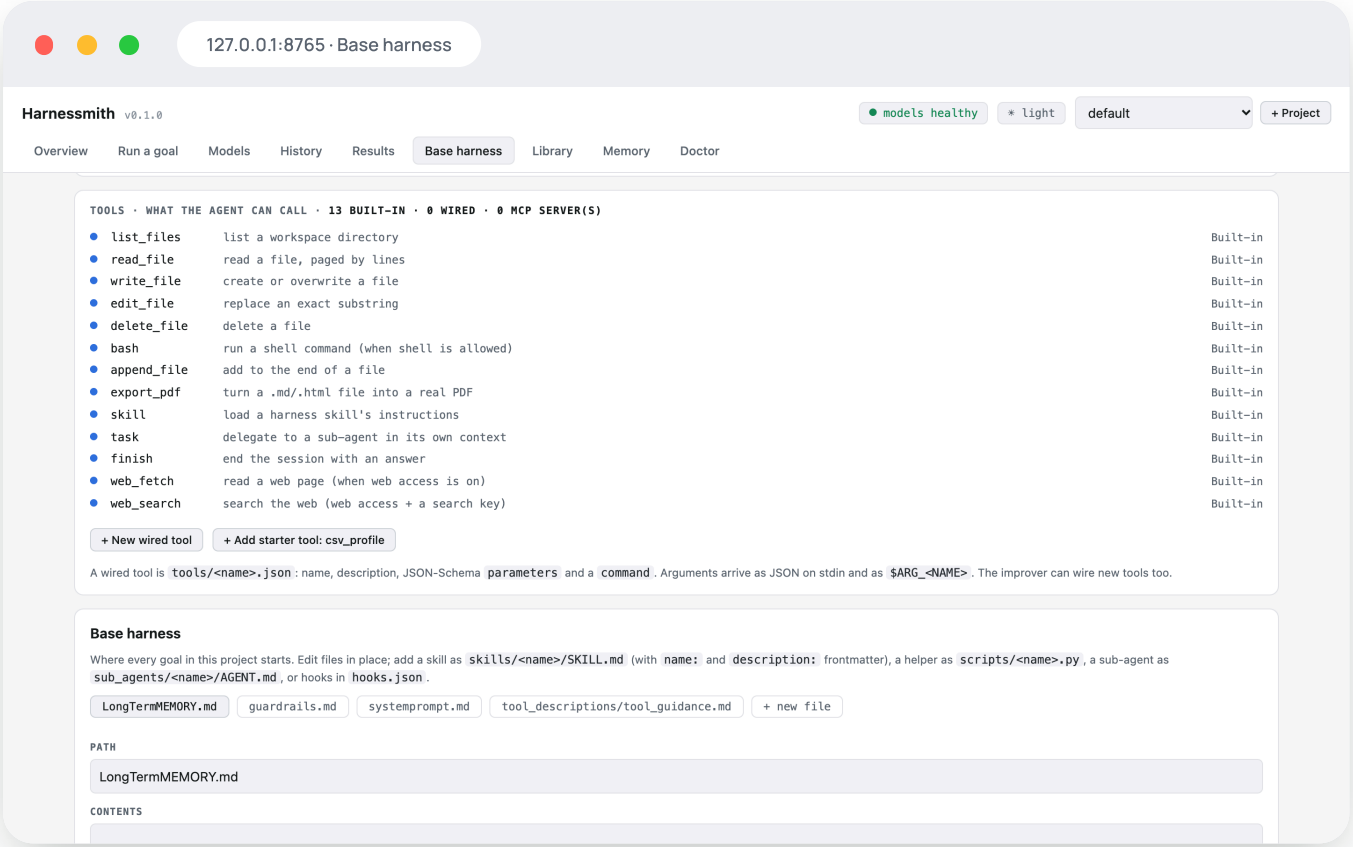


LESSONS THAT CARRY OVER

Lessons for later goals

After each run a reflection writes a lesson: the cue, the lesson, the evidence and a confidence. Later goals in the same domain read the library before proposing edits.

- Pin a lesson you trust: it is listed first for later improvers
- Retire one that no longer helps: it is moved aside, not deleted

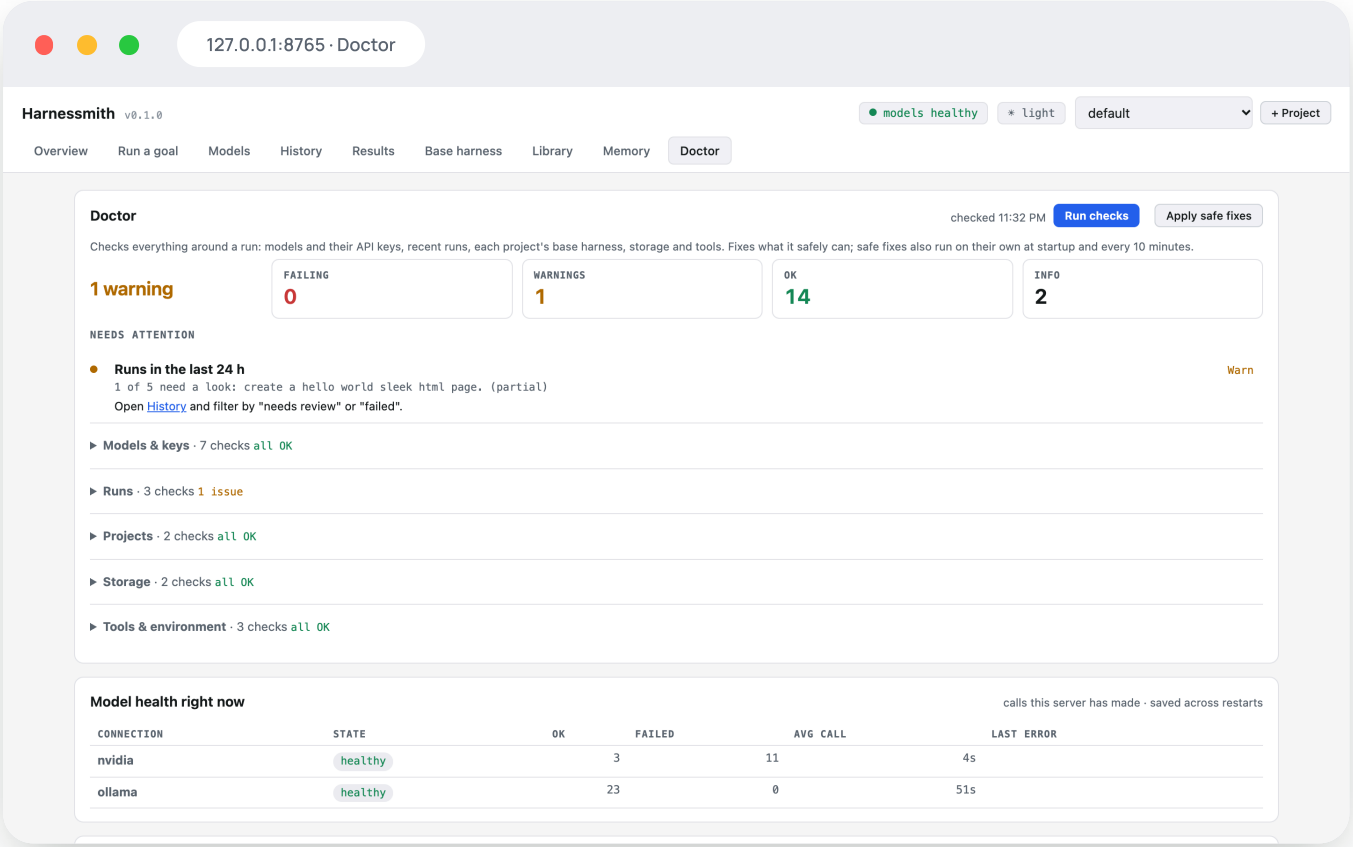


WHAT THE AGENT CAN CALL

What the agent can call

The base harness is where every goal in a project starts: system prompt, guardrails, memory, tool guidance, skills, scripts and wired tools.

- export_pdf turns Markdown or HTML into a real PDF (headless Chrome when installed, a built-in writer otherwise)
- web_fetch and web_search when web access is on for the project
- Wired tools are JSON files under tools/ that run a command; the improver can add them too



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CHECKS AND SAFE FIXES

Checks and safe fixes

The Doctor leads with a summary and lists only what needs attention, each with a next step. The rest is grouped and folded.

- Models & keys: every role's model reachable, the model exists, breaker state, API key present
- Runs: failed or needs-review runs in the last 24 hours, stuck or interrupted runs
- One-click fixes: reset a breaker, resume runs, restore a base harness, make .env private
- Safe fixes also run at startup and every 10 minutes

HARNESSMITH

Give it a goal

That is Harnessmith. Give it a goal, connect a model, and let it improve its own harness. It runs on your machine, and your keys stay in your own dot env file.

Harnessmith

Thank you

This guide covered every major screen in Harnessmith, from describing a goal and connecting a model to reading the results, the lessons it learned, and the Doctor's checks. Keys stay in your own .env file; nothing in this guide shows one.

GIVE AN AGENT A GOAL. WATCH IT IMPROVE ITS OWN HARNESS.